

Akash Kothari

905-226-8698 | akash@kotharigroup.com | akashkothari.ca | linkedin.com/in/akashkothari07 | github.com/akashkothari2007

EDUCATION

University of Waterloo

Bachelor of Applied Science in Computer Engineering (Co-op)

Waterloo, ON

GPA: 4.0/4.0

EXPERIENCE

Software Engineer Intern (Incoming)

Sep. 2026 - Dec. 2026

Shopify

Software Engineer Intern

May. 2026 - Present

QuiikMart - Grocery Delivery Startup

Remote

- Built a recommendation engine with a **nightly sweeper** that rolls purchase totals into a baseline popularity store, layered with per-user signals and **Redis** caching for guest sessions.
- Built a bulk CSV product-upload system processing up to **1,000** products per upload, with server-side validation, flexible column aliasing, category matching, and a two-step validate-then-import flow.
- Wrote **80+** unit and integration tests, validated features in staging environments, and shipped bug fixes across a NestJS/TypeScript and PostgreSQL codebase.

ML Software Engineering Intern

Jan. 2026 - Apr. 2026

RamSoft

Toronto, ON

- Built a real-time clinical transcription pipeline running **MedASR** (medical dictation) and **WhisperFlow** (voice commands) in parallel, fusing both streams via timestamp alignment and fuzzy word matching.
- Engineered chunked streaming, PyTorch word-confidence scoring, and a replay simulator for evaluation, cutting error rate by **80%** across **200** clinical audio samples; deployed to **Kubernetes** via Azure Container Registry.
- Built a multi-step **LLM-as-judge** evaluation system scoring outputs against reference transcripts.
- Designed a **Cosmos DB** → **SQL Server** schema migration for a scheduling epic, with normalized appointment and reservation tables, wall-clock time storage, and full recurrence support.
- Built an internal **MCP server** for Azure SQL with MFA token auth, enabling natural-language database queries.

Cloud Software Engineering Intern

Jun. 2025 - Sep. 2025

Tripleview Technologies

Mississauga, ON

- Automated Teams call-recording retrieval with Azure, Microsoft Graph, and C#, removing **10+ hours/week** of manual work.
- Built an AI summarization pipeline (webhook triggers, polling) that turned **8+ transcripts/hour** into structured **Zendesk** tickets, shipped to production and adopted across internal teams.

PROJECTS

Mercury Mortgages - Expense Automation | React, FastAPI, Supabase, GPT-4o Vision, Redis

- Built a full-stack expense automation platform processing **\$1.4M+** in receipts across **18** sub-companies, auto-matching **1000+** receipts to transactions via a scoring algorithm and replacing a manual bookkeeping role.
- Engineered a dual-ingestion pipeline auto-processing **415+** emails via Microsoft Graph webhooks, using **GPT-4o Vision** to extract receipt fields and threaded workers for async PDF generation and storage.
- Built a rules engine mapping vendor and location data to GL codes across subsidiaries, with a multi-stage report approval workflow, Microsoft SSO, and role-based access control.

FocusOS - Personal Productivity OS | FastAPI, PostgreSQL, React, Azure OpenAI, Docker



- Built a full-stack productivity OS with an AI job-analysis pipeline (**Azure OpenAI**) that scores resume-to-job match, extracts ATS keywords, and generates tailored LaTeX resumes.
- Designed a background task pipeline with non-blocking AI calls, retries, and status polling on **Docker** and **PSQL**.

Self-Driving Car Simulator | Unity, C#, PyTorch, Reinforcement Learning



- Built a vehicle physics engine from scratch in Unity (no built-in wheel colliders): raycast spring-damper suspension, slip-based tire grip via tunable friction curves, and a configurable FWD/RWD/AWD drivetrain.
- Structured the sim as a reinforcement learning environment (raycast sensors, waypoint-based rewards) and am building a **custom PPO training loop from scratch** (no ML-Agents), architected for GPU-parallel training across thousands of environments.

TECHNICAL SKILLS

Languages: Python, C/C++, C#, TypeScript, JavaScript, Java, SQL

Frameworks & Libraries: FastAPI, NestJS, Node.js/Express, Flask, React, Next.js, Three.js, Unity

Cloud & DevOps: Azure, Docker, Kubernetes, PostgreSQL, Redis, Supabase, CI/CD, Git, REST APIs

ML & Data: PyTorch, NumPy, Pandas, Reinforcement Learning, Feature Engineering, MCP, LLM, Evaluation